Duration: 105 minutes | Boğaziçi University | Name:
Nov 11, 2026; 17:00-18:45 | Math 202 Midterm Exam | ID #:
| | | | | | Math #:

Question:	1	2	3	4	Total
Points:	18	18	18	18	72
Score:	A SOLUTION KEY				

• Any electronic device which is visible around you during the exam will be considered as a reason to receive an **F**. No excuses accepted! • You must write your final result in the boxes provided. If a box is empty, that question will not be evaluated!

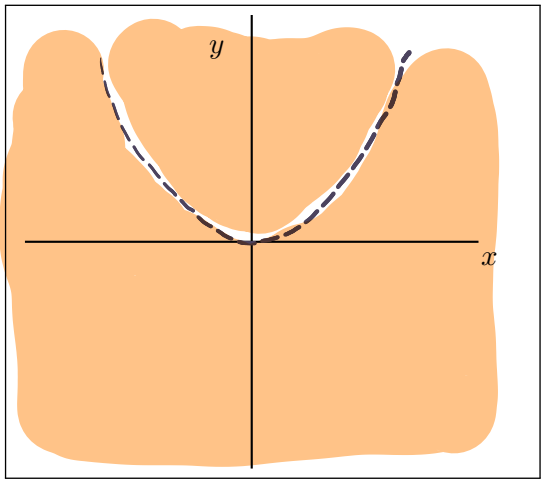
[18] 1. **In this question, no explicit computation is needed**. Anything written out of the boxes will not be evaluated. (Each part is worth 6 points.)

Consider the initial value problem

$$y' = 2x + \sqrt{|y - x^2|}, \quad y(a) = b.$$

Define $D = \{(a, b) \in \mathbb{R}^2 |$ there **exists** a **unique** solution for the IVP around $(a, b)\}$. Sketch D.

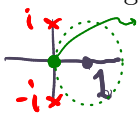
(a)



Check continuity of f & $\partial f/\partial y$ in $y' = f(x,y)$.
f is cont everywhere. $\partial f/\partial y$ is defined and cont when $y \neq x^2$.
This is the existence & uniqueness theorem.

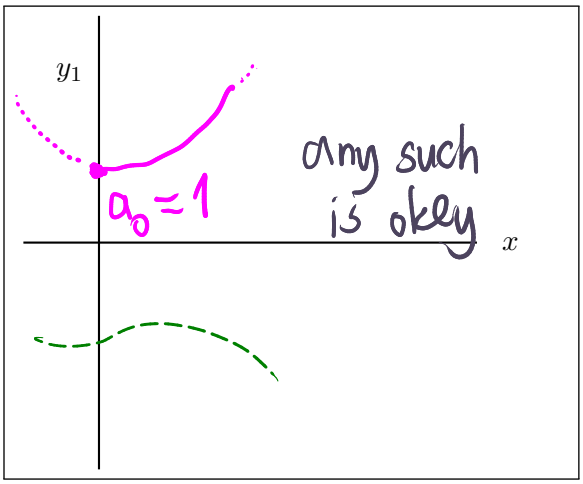
(b) For the ODE $(\ln(1 + x^2))u'' + \frac{1}{2}u' + u = 0$, let u_1 and u_2 be two analytic, independent solutions centered at $x = 1$. Write real numbers in the boxes as **sharp** as possible. (Sharp means "without solving the ODE, I cannot tell something better.")

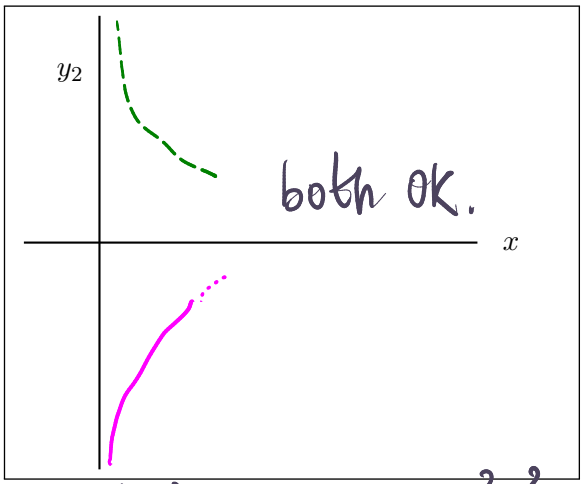
$\frac{1}{\ln(1+x^2)}$, centered at $x=1$, is analytic with $\rho = 1$



$\rho(u_1) \geq$ [1], $\rho(u_2) \geq$ [1]

(c) For the *Laguerre ODE* $xy'' + (1 - x)y' + x^2y = 0$, the point $x = 0$ is a regular singular point. For $x > 0$ let $y_1 = x^r \sum_{k=0}^{\infty} a_k x^k$ be a solution and y_2 be another solution that is independent from y_1. Sketch the behavior of y_1 and y_2 as $x \to 0^+$.





Indicial eqn: $r(r-1) + b_0 r + c_0 = 0$ with $b_0 = \lim_{x\to 0} \frac{x(1-x)}{x} = 1$; $c_0 = \lim_{x\to 0} \frac{x^2 \cdot x^2}{x} = 0$

$r_1 = 0$: $y_1 = x^0 \cdot \left(1 + \sum_{1}^{\infty} a_k x^k\right)$: a power series ; $r_2 = 0$: $y_2 = y_1 \ln x + x^0 \cdot \sum_{1}^{\infty} \ldots$ (explodes)

[18] 2. Solve the initial value problem:

$$\underbrace{(9x^2+y-1)}_{M}-\underbrace{(4y-x)}_{N}\frac{dy}{dx}=0, \qquad y(1)=0.$$

Write down the solution so that y is given **explicitly** (not implicitly) as a function of x.

Since $\frac{\partial M}{\partial y}=1=\frac{\partial N}{\partial x}$, this is an exact ODE.

That is, a potential fnc $\varphi(x,y)$ exists.

$M=\frac{\partial \varphi}{\partial x} \Rightarrow \varphi(x,y)=3x^3+xy-x+C(y)$

$N=\frac{\partial \varphi}{\partial y}=x+C'(y)=x-4y \Rightarrow C(y)=-2y^2+C, \quad C\in\mathbb{R}$

Thus the ODE reads: $\frac{d\varphi}{dx}=0$

$\varphi(x,y)=$ constant solves y implicitly in terms of x:

$\varphi=3x^3+xy-x-2y^2=a$ with $\varphi(1,0)=0$

So $a=3-1=2$ and $2y^2-xy+(x+2-3x^3)=0$

$\Rightarrow y=\frac{1}{4}\left(x-\sqrt{x^2+8(x+2-3x^3)}\right)$

↳ so that $y(1)=0$.

$$y(x)=\boxed{\frac{1}{4}\left(x-\left(x^2+8(x+2-3x^3)\right)^{1/2}\right)}$$

[18] 3. Use **the method of reduction of order** to find a second solution $y_2(t)$ of the ODE below, given that a first solution is $y_1(t) = t$. (No other method is allowed. **Do not use memorized formulas.** Instead deduce those formulas step-by-step here.)

$$t^2y'' + 2ty' - 2y = 0, \quad t > 0.$$

Set $y = u \cdot y_1$: $y' = u'y_1 + uy_1'$, $y'' = u''y_1 + 2u'y_1' + y_1''u$.

Insert & simplify:

$$t^2(u''y_1 + 2u'y_1') + 2tu'y_1 = t^3u'' + 4t^2u' = 0$$

So for $t \neq 0$: $u'' = -\frac{4}{t}u' \Rightarrow u' = \exp\left(-\int \frac{4}{t}dt\right) = t^{-4}$

Hence $u = -\frac{1}{3}t^{-3}$ & $y_2 = -\frac{1}{3}t^{-2}$.

$y_2(t) =$ $\boxed{-1/3t^2}$

4. Consider the ODE $x^2y'' + xy' + (x^2 - 1)y = 0$.

[2] (a) Show that $x = 0$ is a regular singular point.

$\lim_{x\to 0} x \cdot \frac{x}{x^2} = 1 = b_0$

$\lim_{x\to 0} x^2 \cdot \frac{x^2-1}{x^2} = -1 = c_0$

Both limits exist; so $x=0$ is a regular singular point.

[2] (b) Find the roots of the indicial equation.

$F(r) = r(r-1) + b_0r + c_0 = r^2 - 1 = 0 \Rightarrow r_1 = +1,\ r_2 = -1$

[10] (c) Produce a solution for the ODE for $x > 0$ with a power series portion **centered at 0**.

Let $y = x^r \cdot \sum_{k=0}^{\infty} a_k x^k$ & assume the power series portion is analytic.

Then $xy' = \sum_0^\infty a_k (k+r) x^{k+r}$, $x^2 y'' = \sum_0^\infty a_k (k+r)(k+r-1) x^{k+r}$

$$L[y] = \sum_0^\infty \left\{ a_k \left[(k+r)(k+r-1) + (k+r) - 1 \right] x^{k+r} + a_k x^{k+r+2} \right\} = 0$$

$\Rightarrow$ x^r : $a_0 F(r) = 0$

x^{r+1} : $a_1 F(r+1) = 0 \Rightarrow a_1 = 0 = a_3 = a_5 = \cdots$

$(k \geq 1)$ x^{r+k} : $a_k \cdot F(k+r) + a_{k-2} = 0 \Rightarrow a_k = \frac{-1}{F(k+r)} a_{k-2}$, $k \geq 2$

$F(k+r) = (k+r-1)(k+r+1)$

$a_2 = \frac{-1}{(1+r)(3+r)} a_0$, $a_4 = \frac{(-1)^2}{(3+r)(5+r)} a_2 = \frac{(-1)^2 a_0}{(1+r)(3+r)^2(5+r)}$,

$a_{2\ell} = \frac{(-1)^\ell a_0}{[2 \cdot 4 \cdot \cdots \cdot (2\ell+2)]^2} = \frac{(-1)^\ell}{2^{2\ell+2} \cdot (\ell+1)!]^2} a_0 = \frac{(-1)^\ell}{4^\ell \ell! (\ell+1)!} a_0$

$r = 1$, $2 \cdot (2\ell+2)$

(this does not work for the smaller index $r_2 = -1$)

So for $a_0 = 1$: $y(t) = x^1 \cdot \sum_{\ell=0}^{\infty} \frac{(-1)^\ell}{4^\ell \ell! (\ell+1)!} x^{2\ell}$

[4] (d) Determine the radius of convergence of the power series portion of the solution in the previous part.

Since $p(x \frac{Q}{P}) = +\infty$ & $p(x^2 \frac{R}{P}) = +\infty$, by Frobenius thm (without even solving (c)). $\rho = +\infty$

Or you may employ the Ratio Test for your solution in part (c).

End of paper